

ultima\_cluster 2.13.0

# Command cheatsheet

The commands that you use with ultima\_cluster (UC). Part 1 is for developers. Part 2 is for operators. Each entry tells you what a command does and why you use it.

## HOW TO READ THE ENTRIES

- `<DIR>` is the instance directory of a node. `<ID>` is the application identity ( `app_id` ) of the cluster.
- In the `uc2ctl` entries, `...` stands for `--instance-dir <DIR> --app-id <ID>`.
- If the `[admin]` policy of the node is `auth = "hmac"`, add `--admin-key <PATH>` to each `uc2ctl` command that changes the cluster.

## PART 1 · DEVELOPER

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## PART 2 · OPERATOR

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## 1.1 Build an application on UC

### WHAT IT DOES

### WHY YOU USE IT

```
cargo run -p counter --bin counter-single
```

Starts one node and one counter service in one process.

See UC work before you set up a cluster.

```
packaging/quickstart-local.sh --bin-dir target/release
```

Starts three nodes, three services and three gateways on this host, then prints `PASS`.

Make sure that a local build runs end to end. Add `--keep` to leave the cluster up.

```
examples/kv/scripts/kvcluster.sh up --fresh
```

Starts a three-node cluster of the key-value store example on this host.

Use a real application with snapshots and sessions as a model for your own.

```
kv --gateways <HOSTS> put <KEY> <VALUE>
```

Sends one command to the key-value store through a gateway.

Try commands, linearizable reads and retries by hand.

```
cargo test (in your service crate)
```

Runs the unit tests of your FSM.

Test `apply` as a plain function. It needs no cluster.

```
uc2-diffreplay corpus export ... --row <R> --from <P> --out <CORPUS>
```

Copies a snapshot at position P and the log after it from a node into a corpus directory.

Get real input for a replay. Use `--around <POS>` to cut a corpus around a bug.

```
uc2-diffreplay upgrade --corpus <C> --old <BIN> --new <BIN> --declare intent.toml --report <FILE>
```

Replays the corpus through two builds and compares all outputs with your declared intent.

Prove that a new version changes only what you declared. Exit 0 means that you declared each difference.

```
uc2-diffreplay determinism --corpus <C> --bin <BIN> --report <FILE>
```

Replays the corpus through the same build two times.

Find nondeterminism in your FSM. The result must be empty.

```
uc2-diffreplay reconstruction --corpus <C> --bin <BIN> --report <FILE>
```

Compares the state from the snapshot with the state from a replay from the start of the log.

Find a state that depends on how a node recovered.

```
uc2-diffreplay pin-verify --corpus <C> --old <BIN> --new <BIN> --app-id <ID> --fsm <NAME> --to <VER> --report <FILE>
```

Does a pinned upgrade on a live single-node cluster with your real binaries.

Rehearse an upgrade before you do it in production.

## 1.2 Work on UC itself

The contributor command cheatsheet ( `docs/reference/contributor-cheatsheet.md` ) lists the commands that you use to build, test and prove UC itself.

## 2.1 Run the processes

### WHAT IT DOES

### WHY YOU USE IT

```
uc2-node --config /etc/uc2/node.toml
```

Starts a node. Each configuration error stops the start with an error that names the key.

Run one cluster member.

```
uc2-gateway --config /etc/uc2/gateway.toml
```

Starts the TCP gateway for remote clients on the host of a node.

Let clients on other hosts reach the cluster.

```
sudo systemctl enable --now uc2-node
```

Starts the node as a systemd service. Use `uc2-gateway` for the gateway.

Let systemd restart a node after a failure.

```
sudo systemctl stop uc2-node
```

Sends `SIGTERM`. The node drains and stops.

A planned stop. The restart then replays only the tail of the journal.

```
uc2-node --version
```

Prints the version. `uc2ctl` and `uc2-gateway` also take `--version`.

Make sure that each host runs the same release.

## 2.2 Inspect a cluster

### WHAT IT DOES

### WHY YOU USE IT

```
uc2ctl status ...
```

Prints the role, term, positions, members and FSMs of one node.

The first command when you examine a node.

```
curl -s http://<METRICS_BIND>/readyz
```

If the node can serve, it returns 200. If not, it returns 503. `/healthz` tells you whether the node is alive.

Use these for load balancers and health checks.

```
curl -s http://<METRICS_BIND>/metrics
```

Returns all metrics in Prometheus text format.

Examine one metric by hand. Prometheus uses the same endpoint.

```
uc2ctl schedule show ...
```

Prints the committed schedule table.

Make sure that a `schedule apply` took effect.

```
uc2ctl settings show ...
```

Prints the committed replicated settings.

Make sure that a `settings apply` took effect.

## 2.2 Inspect a cluster (continued)

### WHAT IT DOES

### WHY YOU USE IT

```
uc2ctl snapshot show ...
```

Prints the newest snapshot position of each row and of the cluster artifact, and the position of the newest complete set.

Find the position P for a pin, a fetch or a backup.

```
uc2ctl upgrade show ...
```

Prints the upgrade pins in the committed cluster artifact.

Examine the pin history of each row. The live pin is in `uc2ctl status`.

```
uc2ctl audit --instance-dir <DIR> --tail 20
```

Prints the last 20 records of the admin audit log. Add `--json` for raw records.

Find who changed the cluster, and when.

## 2.3 Change the membership

The cluster accepts one membership change at a time.

### WHAT IT DOES

### WHY YOU USE IT

```
uc2ctl add-learner ... --id <N> --addr <IP:PORT>
```

Adds a node that gets the log but does not vote.

The first step to add a node. A learner catches up before it votes.

```
uc2ctl promote ... --id <N>
```

Changes a learner into a voter.

Make a caught-up learner count toward the quorum.

```
uc2ctl demote ... --id <N>
```

Changes a voter into a learner.

The first step to remove a voter.

```
uc2ctl remove-learner ... --id <N>
```

Removes a learner from the cluster.

Take a node out of the cluster. A removed id cannot join again.

```
uc2ctl remove-voter ... --id <N>
```

Removes a voter from the cluster.

Take a voter out in one step, for example the leader.

## 2.4 Change replicated configuration

### WHAT IT DOES

### WHY YOU USE IT

```
uc2ctl schedule apply <FILE.toml> ...
```

Replaces the schedule table of recurrent ticks for all nodes.

Run periodic or daily work in an FSM.

```
uc2ctl settings apply <FILE.toml> ...
```

Changes the four replicated settings for all nodes.

Change `fsm_lag`, `admission_bytes` or the snapshot cadence without a restart. [Configuration § `[settings]`] (configuration.md#settings)

## 2.5 Take snapshots

### WHAT IT DOES

### WHY YOU USE IT

```
uc2ctl snapshot ...
```

Puts a snapshot instant on the log. Each node makes a snapshot of each row at one position.

Bound the growth of the journal, make an upgrade origin, or make a backup point.

```
uc2ctl snapshot ... --standby
```

Only the learners make the snapshot.

Take a snapshot and never stop commit.

```
uc2ctl snapshot fetch ... --from <LEARNER>
```

Copies a complete snapshot set from a learner to this node.

Give a voter a set after a standby instant.

## 2.6 Upgrade

### WHAT IT DOES

### WHY YOU USE IT

```
scripts/uc2_flag_day.sh --hosts ... --upgrade-cmd '...' --yes-traffic-stopped
```

Stops all nodes, runs your upgrade command on each host, starts all nodes and prints the downtime. If a step fails, it starts all nodes again.

Upgrade UC across a change of wire or cnc version.

```
uc2ctl upgrade pin ... --row <R> --to <VER> --origin <P>
```

Records on the log that row R changes to version VER from the snapshot at P.

Make each instance of the new FSM version start from the same state.

## 2.7 Back up and recover

### WHAT IT DOES

### WHY YOU USE IT

```
uc2ctl backup --instance-dir <DIR> --out <DIR>
```

Copies a node's journal, state and snapshots to a new directory. The node can run.

Keep a copy off the node before an upgrade or for disaster recovery.

```
uc2ctl verify-backup <ARTIFACT>
```

Makes sure that a backup is complete and consistent.

Examine a backup before you need it.

```
uc2ctl restore <ARTIFACT> --instance-dir <DIR>
```

Writes a backup into an empty instance directory.

Rebuild a node on a new host.

```
uc2ctl force-single-member --instance-dir <DIR> --app-id <ID> --node-id <N> --confirm-cluster <ID>
```

Makes one stopped node the only member of its cluster.

Recover after the cluster loses a majority of its voters for ever. The cluster can lose committed writes.

## 2.8 Secure the admin commands

### WHAT IT DOES

### WHY YOU USE IT

```
uc2ctl gen-admin-key <PATH>
```

Writes a new 32-byte admin key file with mode `0600`.

Sign admin commands when `[admin] auth = "hmac"`.